Unit A2: Networks (HL Extension)

Sub-topics: A2.1.5, A2.2.2, A2.3.4, A2.4.2, A2.4.3

1. Syllabus Scope & Learning Outcomes

These topics are for Higher Level students only.

Syllabus Point	Command Term	Student Expectation (The "Do")
A2.1.5	**Describe**	Describe the function of the TCP/IP Model layers: Application, Transport, Internet, Network Interface.
A2.2.2	**Describe**	Describe the function of servers: DNS, DHCP, File, Mail, Proxy, Web.
A2.3.4	**Explain**	Explain Static vs. Dynamic Routing (tables, updates, fault tolerance).
A2.4.2	**Describe**	Describe network vulnerabilities: DDoS, MitM, Phishing, SQL Injection, XSS, Zero-day.
A2.4.3	**Describe**	Describe network countermeasures: CSP, MFA, IDS/IPS, SSL/TLS, Updates.

2. Key Terminology & Definitions

- **TCP/IP Stack:** The conceptual model of networking (Application -> Transport -> Internet -> Physical).
- **DNS (Domain Name System):** Translates human-readable names (google.com) to IP addresses.
- **DHCP (Dynamic Host Configuration Protocol):** Automatically assigns IP addresses to devices joining a network.

- **Proxy Server:** Acts as an intermediary for requests from clients (caching, filtering).
- **Reverse Proxy:** Acts on behalf of servers (load balancing, security).
- **DDoS (Distributed Denial of Service):** Overwhelming a target with traffic from multiple compromised sources (botnets).
- **SQL Injection:** Inserting malicious SQL queries into input fields to manipulate a database.
- **Phishing:** Deceptive communications (emails) to steal credentials.
- **Zero-day Exploit:** An attack on a software vulnerability that is unknown to the vendor (no patch exists yet).

3. Core Content & Exam Actions

A2.1.5: TCP/IP Model

Theory:

- **Application Layer:** Where protocols like HTTP, DNS, SMTP reside. User interaction.
- **Transport Layer:** TCP/UDP. Handles segmentation, error checking, and ports.
- **Internet Layer:** IP. Handles addressing and routing packets across networks.
- **Network Interface Layer:** Physical transmission (Ethernet, WiFi, MAC addresses).

Student Exam Action:

- **Mapping:** Be able to place protocols into their correct layer (e.g., HTTP = Application, TCP = Transport, IP = Internet).

A2.2.2: Servers

Theory:

- **DNS:** Hierarchical/distributed database. Critical for usability.
- **Proxy:**
  - **Forward Proxy:** Hides the client, filters content (schools use this), caches web pages to save bandwidth.
  - **Reverse Proxy:** Hides the server, balances load among multiple servers, handles SSL encryption.
- **DHCP:** "Leases" IP addresses. Process: Discover -> Offer -> Request -> Acknowledge (DORA).

Student Exam Action:

- **Explanation:** Explain why a company would use a **Reverse Proxy** (to protect internal servers and handle high traffic loads).

A2.3.4: Routing

Theory:

- **Static Routing:** Routes manually configured by admin.
  - *Pros:* Secure, predictable, low CPU usage.

    - *Cons:* Does not adapt to failures, high maintenance.
- **Dynamic Routing:** Routers talk to each other to update tables.
    - *Pros:* Fault tolerance (finds new paths), scalable.
    - *Cons:* Higher CPU/Bandwidth usage, slower convergence.

Student Exam Action:

- **Comparison:** Compare Static vs. Dynamic routing for a small office (Static is fine) vs. an ISP (Dynamic is essential).

A2.4.2 & A2.4.3: Vulnerabilities & Countermeasures

Theory:

- **Attack: SQL Injection** -> **Countermeasure:** Input Validation (sanitizing user inputs).
- **Attack: Phishing** -> **Countermeasure:** Training, Email Filtering, MFA (Multifactor Authentication).
- **Attack: DDoS** -> **Countermeasure:** Rate Limiting, Traffic Analysis, Cloud Mitigation.
- **Attack: Man-in-the-Middle** -> **Countermeasure:** Encryption (HTTPS/VPN).

Student Exam Action:

- **Evaluation:** Evaluate the effectiveness of a Firewall against an SQL Injection. (Low effectiveness—firewalls block ports, but SQL injection happens *inside* allowed web traffic (Port 80/443). Input validation is the real cure).

4. Common Pitfalls & Examiner Tips

- **Pitfall (Layers):** Confusing the **Transport** layer (ensuring data integrity/sequencing) with the **Internet** layer (getting packets to the right address).
- **Pitfall (Proxies):** Confusing Forward and Reverse proxies. Forward = acts for Client. Reverse = acts for Server.
- **Tip (Security):** "Install Anti-virus" is rarely a sufficient answer for HL questions. Be specific: "Use Input Validation," "Implement MFA," "Configure IDS/IPS."